



AL-2001 Artificial Intelligence Lab # 5

Objectives:

- File handling, Regular Expression

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your logic on Copy / Notebook.
2. Write **Your Name** and **Roll No** on your Paper/Sheet's first page.
3. **Do not copy from any source otherwise you will be penalized with negative marks.**
4. Complete your lab **within given Time Slot**.
5. Paste all your codes along with screenshots in a word file and renamed with your roll number.
6. Keep all your source files in your computer for verification. Do not overwrite a single source file for all programs.

Problems:

1. You're developing a registration system for a website, and you want to ensure that users create strong passwords to protect their accounts. Write a program to check the strength of a password entered by a user.
2. You are working on a code analysis tool, and you want to find and extract variable names from a given text. Variable names in the code can have different patterns, but they typically consist of letters, numbers, and underscores. Write a program to find and display variable names in the text file with the combination of wildcards and regular expressions.
3. You are tasked with creating a Python program that extracts the roll numbers of students from their email addresses. Students have email addresses that follow a specific pattern, which includes their roll numbers. The email addresses are provided in a text file. Your program should read the file, extract the roll numbers using regular expressions, and display them in a properly formatted list.
4. You are developing a Python program to calculate spam confidence scores for a set of text messages. The spam confidence score is a measure of how likely a message is to be spam based on specific patterns and keywords often found in spam messages. You will use regular expressions and wildcards to identify these patterns and assign scores accordingly.

Program Steps:



Define a set of regular expressions with wildcards that match common spam patterns, such as words in all capital letters, excessive use of exclamation marks, or offers for products or services.

Read a collection of text messages from a file or user input.

For each message, apply the defined regular expressions to identify spam patterns.

Assign a spam confidence score to each message based on the number of matches to the spam patterns.

Display the spam confidence scores and indicate whether each message is likely spam or not.

You need to do your exercise within the given time.